SmartMotion: Project Brief and Claude Code Kickoff

Owner: Sam Jones
Prepared: 7 November 2026
Purpose: This file hands the SmartMotion project over to a Claude Code session working in a new GitHub repository. It summarises everything decided and researched so far, and tells Claude Code how to set up the repo and build a **living dashboard** that tracks every task, contact, document and decision needed to take SmartMotion live as a business.

> **Claude Code: read the whole file first, then follow Section 12 ("Instructions for Claude Code").** Treat this brief as the starting knowledge base. Sam will add more documents over time (3D printing work, engineering files, business documents). Fold each one into the knowledge base and the dashboard as it arrives.

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1. What SmartMotion is

SmartMotion is a UK business built around a **6-DOF (six degrees of freedom) full-motion simulator** that Sam designed and built. The rig turns flight and driving game software into real movement. Riders see the action through a VR headset or screens and use force-feedback joysticks or steering wheels.

- **Website:** https://www.smartmotion.tech (product info: https://www.smartmotion.tech/product-information)
- **Public contact:** info@smartmotion.tech · 07968 615044
- **Currently presented as:** "A Skysmart MRO Ltd Company", based in Great Dunmow, Essex (about 5 minutes from Stansted Airport). Skysmart's own site links to it at https://www.skysmart.co.uk/flight-simulator.
- **Plan:** launch SmartMotion properly as a business in 2027.

Two business lines

1. **Experiences (first priority, launching 2027).** Paid sessions on the rig at the Great Dunmow base, at corporate events and at airshows and other events. Gift vouchers go on sale first.
2. **Building and selling simulators (later).** SmartMotion intends to develop and sell its simulators, with options from complete systems (PC and VR installed) to kit packages. The website already says rigs are built "to suit your budget". During this chat, rig sales were explicitly **parked until the experiences business is running**. They stay on the roadmap but get no launch tasks yet.

A third area, **3D printing**, will be added. Sam will provide documents on what he does with 3D printing and how it fits (for example parts for rigs, cockpit controls, merchandise). Create a placeholder for it in the knowledge base and dashboard.

History and track record (public sources)

- **MRO Europe 2024:** the simulator started as a "bit of fun" for Skysmart's stand and ran a "Top Gun" low-level-flying time trial (an F/A-18E Super Hornet through a valley). Aviation Week reported that it proved popular enough to become a new business line: https://mroevents.aviationweek.com/articles/exhibition-gimmick-spawns-new-business/
- **The idea came from** a local Scouts visit to Skysmart, where a gliding club set up a basic simulator. Sam then spent about five months adding the motion rig and VR.
- **2025:** Duxford Summer Airshow (Sally B fundraiser), Shuttleworth and Wycombe Air Park.
- **2026:** Duxford several times. A Skysmart STEM and Aviation Day was held at IWM Duxford in June 2026.
- **MRO Europe, Amsterdam, 27–29 Oct 2026:** a next-generation simulator was unveiled, with a Low Level Flight Challenge on the stand. Returning visitors came back to compete. The rig travelled by ferry (Harwich to Hook of Holland) under an ATA Carnet arranged through Maltacourt.
- **Story assets:** Skysmart owns a real ex-RAF **Scottish Aviation Bulldog** trainer. This is a strong marketing hook ("fly the Bulldog in VR").

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2. The product (technical summary)

Item	Detail
Platform	Single-seat 6-DOF motion platform: pitch, roll, yaw, surge, heave, sway
Actuators	7 electric actuators: 6 for motion, plus 1 driving seat-belt tensioners that simulate g-force. No hydraulics.
Motion controller	Motion4Sim servo driver controller (upgraded to the newest generation in Oct 2026)
Motion software	FlyPT Mover (telemetry to motion; profiles per game)
Display	VR headset or screens, plus a spectator screen
Controls	Force-feedback joystick and throttle (flight) or steering wheel and pedals (driving); the rig converts between modes
Restraint	5-point harness
Power	Single-phase electric; portable for events

To confirm and record: rated rider load, footprint and exclusion zone, E-stop design, transport weight and size, PC specification, headset model. These values are needed for the ADIPS design review and the operations manual.

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3. Business plan summary

Source: *SmartMotion 6DOF Simulator Experiences — Business Plan* (31 Oct 2026, 22 pages). Sam has the PDF; add it to `docs/source/`.

- **Three sales channels:** bookable sessions at the Dunmow base, corporate days (on-site or at the client) and walk-up sessions at airshows and events.
- **Target:** about **£13,000 a month** revenue within 12 months in the base case, and about £6,400 a month profit before owner pay. Break-even is about £3,900 a month (roughly 84 public sessions or 3 corporate days).
- **Selling point:** one rig does two jobs, full-motion flight in VR and rally or track driving. Rivals are mostly fixed-base flight sims or 2–3-axis racing rigs. Tagline: **"Fly it. Drive it. Feel it."**
- **Priority customers:** corporate team days and gift-voucher buyers first. Then aviation enthusiasts, student pilots and events.
- **Example prices:** £20 for a 10-minute taster, £39 for 20 minutes, £55–£65 for 30 minutes, £109 for a 60-minute Mission Hour and £159 for the flight-plus-car double. Corporate days (ex VAT) are £795 for a team challenge at the base, £1,250 for a half day on-site and £1,950 for a full day on-site.
- **Operations:** fixed flight days and car days (to avoid repeated conversions), 45-minute slots, and utilisation targets of 40%, rising to 60–70%.
- **Start-up costs:** about £9,600–£19,000, excluding software licences.
- **Launch:** full public opening **March 2027**. **Gift vouchers on sale from the end of November 2026** for the Christmas season.
- **Pricing check needed:** the nearest competitor, **VR Sim** in St Ives, Cambridgeshire (4-DOF rigs running DCS and Assetto Corsa), advertises "from £20 for 30 minutes mid-week". Recheck the plan's prices against theirs before vouchers go on sale.

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4. Legal and compliance: what's needed before taking money

Before selling vouchers or tickets, SmartMotion needs:

1. **ADIPS inspection.** The Amusement Device Inspection Procedures Scheme is the UK industry scheme for rides that move members of the public (HSE guidance HSG175). As a new, self-built device the rig needs a **design review**, **assessment of conformity to design** and an **initial test**, after which it gets a **Declaration of Operational Compliance (DOC)**. It then needs an annual in-service inspection. Event organisers will ask for the DOC.
2. **Insurance.** Public and products liability of £5–10m, with paying riders on a moving 6-axis rig named in writing. Also employers' liability, equipment and goods-in-transit cover, and business interruption cover.
3. **Commercial software licences.** Consumer game licences do not allow paid public use (see Section 5).
4. **Booking, payments and vouchers.** An online system plus terms and conditions, a rider declaration and a UK GDPR privacy notice.
5. **Business admin.** Decide whether SmartMotion trades under Skysmart MRO Ltd or as its own company, confirm the VAT position, get landlord consent and check planning use at the Dunmow

unit, and set up a separate bank account or cost centre.

Key point: by 30 Nov 2026 SmartMotion can legally sell **gift vouchers and dated tickets for 2027 sessions**, provided insurance, terms and licences are in place. **No paid rides until the ADIPS DOC is issued.**

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5. Software licensing findings

Software	Use	Commercial route	Cost / status
X-Plane 12 Professional	Flight	Buy direct from x-plane.com. A professional key is required whenever X-Plane is used to generate revenue.	$1,000 digital; $1,250 USB key (USB key works offline)
DCS World	Jets and warbirds	Eagle Dynamics offers a separate commercial site licence. Contact support@digitalcombatsimulator.com. VR Sim (St Ives) holds one, so there's a precedent.	By quote
Assetto Corsa Commercial Pro	Track and rally (rally via mods)	Sold only through authorised resellers such as Vesaro (Kent) and FPZERO. Choose the Pro tier because it allows third-party software like FlyPT Mover.	About £3,750 per simulator, one-off
rFactor 2	Circuit racing	Commercial licence via info@motorsportgames.com	By quote
FlightGear	Flight (open source, GPL)	Free; paid use is allowed	Free; candidate base for a custom SmartMotion sim
FlyPT Mover	Motion control	Confirm commercial use with the developer (XSimulator forum)	To confirm
Microsoft Flight Simulator 2024	n/a	**No commercial licence available.** Do not use for paid sessions.	Not usable
Prepar3D	n/a	Licensed for training, explicitly not for entertainment	Not suitable
Automobilista 2, EA WRC, DiRT Rally	n/a	No commercial licence found	Not usable

Third-party aircraft, scenery and mods each need separate written permission. **Advertise experiences**

by type ("Fast Jet Low-Level", "WW2 Fighter Sortie", "Rally Stage") rather than by game or brand name.

Custom simulator idea (parked): build a SmartMotion-owned flight experience on FlightGear, adding a UDP telemetry bridge into FlyPT Mover, one-click experience presets and an operator panel. A later option is a Unity or Unreal sim using the JSBSim flight model and OpenXR. A four-phase plan was outlined: prove VR and motion work → aircraft and scenery → experience presets → trial runs and paperwork.

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6. Contacts directory

ADIPS inspection bodies (registered for design review, conformity and initial test)

Company	Contact	Email	Phone	Address	Notes
John H Rundle Ltd	Ken Rundle	clare@rundles.co.uk	01205 480431	Main Road, New Bolingbroke, Boston, Lincs PE22 7LN	Nearest full-scope body. **Send first.**
TÜV SÜD	Patrycjusz Idzikowski	patrycjusz.idzikowski@tuvsud.com	03300 169924	Room 103, County Way, Barnsley S70 2AG	Large international firm; credibility with insurers. **Send first.**
Westlakes Engineering Ltd	Ron Dale	attraction-safety@westlakes.co.uk	0161 236 8203	Unit 2.1, 20 Dale Street, Manchester M1 1EZ	**Send first.**
Leisure Technical Consultants	Bob Nicholls	ltci@ltci.co.uk	0161 351 9526	73 Crescent Road, Dukinfield SK16 4EU	Backup; also offers consultancy
Capstone Inspections	Paul Capper	paul@capstone-inspections.com	07736 641420	19 South Road, Dorset BH19 2QR	Backup
R. J. Associates	Craig Kirkman	craig@rj-a.com	07850 445934	Cheshire	Backup
Comply: Amusement	Grace Narca	comply@complysafe.co.uk	07841 658643	Worcestershire	Backup

Safety					
Kraken NDT (Essex)	Jay Wilson	hireamusement@gmail.com	07450 447958	Chestnut Farm, South Road, South Ockendon RM15 6DT	**Testing only** (non-destructive weld/crack testing); local
ADIPS Ltd	n/a	office@adips.co.uk	n/a	n/a	Scheme office; register at adips.co.uk

Source: https://adips.co.uk/find-a-registered-inspection-body/. The register contains some errors, so verify each email on the company's own website.

Insurance brokers

Broker	Email	Phone	Address / notes
Broker2U (Deborah)	deborah@broker2u.co.uk	01245 226318	Chelmsford area; fairground and funfair specialist. **Send first.**
Acer Insurance Services	info@acerinsurance.co.uk	01959 546823	Winterton House, High Street, Westerham, Kent TN16 1AQ; showmen specialist. **Send first.**
Peacock Insurance Services	info@peacockinsurance.co.uk	02476 437 600	1410 Spring Place, Herald Avenue, Coventry CV5 6UB; amusement park cover. **Send first.**
Skysmart's existing broker	n/a	n/a	May be able to extend the current policy. **Send first.**
Livingstones Insurance	web form	020 8903 9412	Fairground operators' cover, public liability up to £10m. Backup.
Arthur J Gallagher	ajg.com/uk	n/a	Theme parks and leisure attraction team. Backup.

Software vendors

- Eagle Dynamics (DCS): support@digitalcombatsimulator.com. Registered address: Route de la Glâne 107, 1752 Villars-sur-Glâne, Switzerland.
- Laminar Research (X-Plane): x-plane.com
- Assetto Corsa resellers: Vesaro (Kent), FPZERO (fpzero.co.uk)
- Motorsport Games (rFactor 2): info@motorsportgames.com

Market

- **VR Sim**, St Ives, Cambridgeshire (vr-sim.co.uk): nearest competitor. Runs 2 × 4-DOF DCS rigs and an Assetto Corsa rig. Prices from £20 for 30 minutes mid-week.
- Other local simulator experiences: VA Airline Training at Cambridge Airport (A320/737 simulators; £360 for 1 hour).

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7. Master task list (status as of 7 Nov 2026)

Target: **legal and ready for voucher sales by 30 Nov 2026**, with the **first paying riders in March 2027**.
Status key: `done · in_progress · todo · waiting · parked`

Already done

ID	Task	Status
D-01	Business plan written (31 Oct 2026)	done
D-02	Research: which games can legally be used commercially	done
D-03	Research: ADIPS process and registered inspection bodies	done
D-04	Research: specialist insurance brokers	done
D-05	Launch checklist written (Docs + PDF)	done
D-06	ADIPS inspection request email drafted	done
D-07	Insurance quote request email drafted (5-point harness confirmed)	done
D-08	Operations manual draft written (11 sections + 3 appendices; Docs + PDF)	done
D-09	Emails pack PDF (both emails + recipients)	done

Week 1 (3–9 Nov): business set-up and long-lead requests

ID	Task	Area	Status
W1-01	Send ADIPS email to Rundle, TÜV SÜD, Westlakes	Safety	todo (confirm if sent)
W1-02	Send insurance email to Broker2U, Acer, Peacock and Skysmart's broker; ask for quotes by about 14 Nov	Insurance	todo (confirm if sent)
W1-		Admin	todo

03	Decide with Andy: SmartMotion as a Skysmart division or a new Ltd		
W1-04	Check the "SmartMotion" name and trademark (UK IPO)	Admin	todo
W1-05	Accountant: VAT position; separate bank account or cost centre	Finance	todo
W1-06	Landlord consent and planning-use check for public visitors	Admin	todo
W1-07	Licence requests: Eagle Dynamics, Assetto Corsa reseller, FlyPT Mover developer; buy X-Plane Pro	Software	todo

Week 2 (10–16 Nov): insurance and safety paperwork

ID	Task	Area	Status
W2-01	Compare insurance quotes; bind public liability cover	Insurance	waiting on W1-02
W2-02	Choose and book the ADIPS inspection body	Safety	waiting on W1-01
W2-03	Written risk assessment (base + event template)	Safety	todo
W2-04	Complete the operations manual [bracketed] values with rig data	Safety	todo
W2-05	Set rider restrictions (age, height, weight, medical)	Safety	todo
W2-06	Gentle, Medium and Strong motion profiles per title	Engineering	todo
W2-07	PAT testing of all electrical kit	Safety	todo

Week 3 (17–23 Nov): licences and booking system

ID	Task	Area	Status
W3-01	Sign at least one flight and one car commercial licence	Software	todo
W3-02	Choose booking system (Bookeo / Checkfront)	Sales	todo
W3-03	Set up vouchers: unique codes, 12-month expiry, redemption tracking	Sales	todo

W3-04	Card payments set up and tested (Stripe / Square / SumUp)	Sales	todo
W3-05	Health and eligibility questions in booking flow	Sales	todo

Week 4 (24–30 Nov): terms, website and go-live

ID	Task	Area	Status
W4-01	Booking and voucher terms (solicitor check)	Legal	todo
W4-02	Rider declaration and parental consent form	Legal	todo
W4-03	UK GDPR privacy notice	Legal	todo
W4-04	Website: Experiences page, prices, booking button, Gift Vouchers page, FAQ	Website	todo
W4-05	Fix the outdated "coming to market in 2025" text and the placeholder launch news post	Website	todo
W4-06	Google Business Profile for the Dunmow base	Marketing	todo
W4-07	Recheck prices against VR Sim	Sales	todo
W4-08	**Go-live gate:** insurance certificate held, inspection date confirmed, licences signed, test purchase passed → open voucher sales and start the Christmas campaign	Launch	todo

Before the first paying rider (Dec 2026 – Mar 2027)

ID	Task	Area	Status
P-01	ADIPS design review, conformity, initial test; DOC issued	Safety	todo
P-02	Operator training and written sign-off	Safety	todo
P-03	Base fit-out: barrier, waiting area, spectator screen, ventilation	Operations	todo
P-04	First-aider, AED plan, hygiene kit, incident log, RIDDOR procedure	Safety	todo
P-			

05	Maintenance log, spares box, spare headset, backup PC image	Operations	todo
P-06	Staff and friends trial sessions; tune profiles for motion sickness	Operations	todo
P-07	Event pack: insurance, DOC, risk assessment, power spec	Events	todo
P-08	Book 2027 airshows and car shows	Events	todo
P-09	Corporate target list (150 organisations within 60 miles), brochure, demo slots	Sales	todo

Parked / future

ID	Task	Status
F-01	Simulator sales line: pricing, kit packages, design certification for buyers	parked
F-02	Custom SmartMotion sim (FlightGear first version)	parked
F-03	3D printing workstream (awaiting Sam's documents)	parked
F-04	Second rig (when weekend utilisation stays above 75% for 3 months)	parked
F-05	Bulldog DCS module or custom aircraft (ask Eagle Dynamics)	parked

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8. Documents created so far

Document	Format	Where
SmartMotion business plan	PDF (Sam has it)	Add to `docs/source/`
SmartMotion Launch Checklist	Claude Doc + PDF	https://claude.ai/artifact/SQ2MhWDLs1xKCTqy4zS7Do · `SmartMotion_Launch_Checklist.pdf`
SmartMotion Operations Manual (draft v0.1)	Claude Doc + PDF	https://claude.ai/artifact/4DUSRs4hFLLmB64XDfLdwz · `SmartMotion_Operations_Manual.pdf`
Inspection and Insurance Emails pack	PDF	`SmartMotion_Inspection_and_Insurance_Emails.pdf`

Sam should copy these PDFs into `docs/source/` in the repo.

9. Open questions and decisions

- Trading entity: a Skysmart MRO Ltd division, or a new SmartMotion Ltd?
- VAT treatment of public prices.
- Final rider limits and rig data for the manual (load, height, age, exclusion zone, E-stop behaviour).
- Which simulation titles to license first (recommended: X-Plane Pro now; ask Eagle Dynamics about DCS; Assetto Corsa Commercial Pro for driving).
- Final prices after the VR Sim comparison.
- Booking system choice.
- When to reopen the simulator sales line.
- How 3D printing fits into the business.

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10. Working principles agreed in this chat

- Experiences first; simulator sales later.
- Nothing paid runs on the rig until ADIPS sign-off and insurance are in place.
- Never charge for sessions on consumer-licensed software.
- Advertise experiences by type, not by trademarked aircraft or game names.
- Keep written records of everything: inspections, checks, training, incidents, licences.

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11. Sources

- SmartMotion website: https://www.smartmotion.tech · https://www.smartmotion.tech/product-information
- Aviation Week, "Exhibition 'Gimmick' Spawns New Business": https://mroevents.aviationweek.com/articles/exhibition-gimmick-spawns-new-business/
- Aviation Week MRO Europe 2024 gallery: https://aviationweek.com/shows-events/mro-europe/gallery-products-services-display-mro-europe-2024
- Air League, Skysmart STEM and Aviation Day (June 2026): https://airleague.co.uk/news/skysmart-stem-and-aviation-day-ignites-scouts-curiosity-in-aviation-pathways
- ADIPS register: https://adips.co.uk/find-a-registered-inspection-body/
- X-Plane licensing: https://x-plane.helpscoutdocs.com/article/25-what-license-do-i-need
- DCS licence: https://www.digitalcombatsimulator.com/en/support/license/
- Assetto Corsa commercial licensing: https://simpullsports.com/guides/assetto-corsa-commercial-license
- DCS forum (VR Sim licence precedent): https://forum.dcs.world/topic/269985-dcs-motion-simulator/

12. Instructions for Claude Code

12.1 Goal

Turn this repository into SmartMotion's **single source of truth**: a knowledge base of Markdown documents plus a **dashboard Sam can open in a browser** showing every task, its status and due date, progress towards the go-live gates, contacts, licences, documents and open decisions. The dashboard must be easy to keep up to date. Whenever Sam reports progress or adds a document, Claude Code updates the data files and the dashboard reflects the change.

12.2 Suggested repository structure

```
/
├── CLAUDE.md              # Short project memory for future sessions (create from this brie
├── README.md              # What the repo is, how to open the dashboard
├── docs/
│   ├── brief.md           # This file
│   ├── business/          # Business plan summary, pricing, financials
│   ├── compliance/        # ADIPS, insurance, licences, legal
│   ├── operations/        # Operations manual (Markdown), checklists, SOPs
│   ├── product/           # Rig spec, software stack, engineering notes
│   ├── 3d-printing/       # Placeholder; Sam will add documents
│   ├── marketing/         # Website copy, campaigns, events
│   └── source/            # Original PDFs and files Sam uploads
├── data/
│   ├── tasks.json         # All tasks (Section 7)
│   ├── contacts.json      # Section 6
│   ├── licences.json      # Section 5
│   ├── documents.json     # Section 8 plus anything added later
│   ├── decisions.json     # Section 9 (open and decided)
│   └── milestones.json    # Go-live gates and key dates
└── dashboard/
    └── index.html         # Reads /data/*.json and renders the dashboard
```

12.3 Data model (suggested)

`tasks.json` : an array of objects:

```
{
  "id": "W1-01",
  "title": "Send ADIPS email to Rundle, TÜV SÜD, Westlakes",
  "area": "Safety",
  "phase": "Week 1",
  "status": "todo",
  "priority": "high",
  "due": "2026-11-09",
  "owner": "Sam",
  "depends_on": [],
```

```
  "notes": "",
  "links": []
}
```

Allowed `status` values: `todo`, `in_progress`, `waiting`, `done`, `parked`. Allowed `area` values: Admin, Finance, Legal, Insurance, Safety, Software, Engineering, Operations, Sales, Website, Marketing, Events, Launch, 3D Printing, Product Sales.

The other files follow the same idea: one object per contact, licence, document, decision or milestone, each with an `id`, `status` and `updated` date.

12.4 Dashboard requirements

- A single self-contained `dashboard/index.html` (no build step), served with **GitHub Pages** so Sam can open it from any device, including his phone.
- **Header:** days until the voucher go-live (30 Nov 2026) and until the first paying riders (March 2027), plus overall % of tasks done.
- **Go-live gates:** insurance held · inspection booked / DOC issued · licences signed · booking system tested. Each shows as done or not done.
- **Task board:** grouped by phase or area, filterable by status, area and owner. Overdue tasks are highlighted, and "waiting on" dependencies are shown.
- **Panels:** contacts (with click-to-email and click-to-call), licences, documents (with links), open decisions and a recent-changes log.
- Mobile-friendly. Sam mostly uses a phone.

12.5 First session: do this

1. Create the folder structure above. Save this file as `docs/brief.md`.
2. Write a concise `CLAUDE.md` with project purpose, conventions, where data lives and how to update the dashboard.
3. Turn Sections 5–9 of this brief into the JSON data files.
4. Build `dashboard/index.html` and give Sam the steps to switch on GitHub Pages.
5. Split the brief into the topic documents under `docs/`.
6. Ask Sam to confirm which Week 1 tasks are already done (for example, whether the ADIPS and insurance emails have been sent) and update the statuses.
7. Ask Sam to upload the business plan, checklist, operations manual and emails PDFs into `docs/source/`.

12.6 Ongoing workflow

- When Sam reports progress, update `tasks.json` (status, notes, date), add an entry to the change log and commit with a clear message.
- When Sam uploads a document, file it in `docs/source/`, summarise it into the right topic doc, and add any new tasks, contacts or decisions to the data files.
- Keep everything factual. Mark assumptions clearly, and never mark a compliance item done without

Sam's confirmation.